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PROJECT REPORT

RetailPulse: Discovering Customer Purchasing Patterns

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1 Introduction

The goal of *RetailPulse* is to use real e-commerce transaction data to understand how customers buy. A raw list of invoices does not tell a retailer who its best customers are, who buys regularly, and who has stopped coming back, and treating all customers alike is wasteful. Our problem is to segment customers based on their purchasing behaviour so that each group can be understood on its own terms. Specifically, we seek to (i) group customers into a small number of segments with similar behaviour, (ii) identify repeat buyers and how they differ from one-time buyers, (iii) describe how spending is distributed across customers and over time, and (iv) summarise each segment in plain terms. The project follows a data-centric workflow of cleaning, feature engineering, unsupervised learning and quantitative evaluation.

2 Proposed Solution

Data preprocessing. We remove records without a customer identifier, cancelled invoices, and records with non-positive quantity or price. The value of each line item is then computed as

$$\text{Line value} = \text{Quantity} \times \text{Unit price}, \tag{1}$$

and aggregated per invoice and per customer.

RFM feature engineering. Each customer is described by three standard measures:

- *Recency (R):* days since the most recent purchase, measured from a reference date just after the last transaction.
- *Frequency (F):* number of distinct invoices.
- *Monetary value (M):* total amount spent.

As frequency and monetary value are typically right-skewed, we apply a log transformation where needed.

Exploratory analysis. We visualize the distributions of R, F and M, sales and orders over time, spending by country, and the revenue share of the top customers, and compare repeat buyers (more than one invoice) with one-time buyers.

Clustering. After standardizing the RFM features, we apply K-Means to partition customers into a small number of segments. The number of clusters k is chosen by comparing the silhouette score and the within-cluster sum of squares over a range of candidate values. The silhouette score of customer i is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}}, \tag{2}$$

where $a(i)$ is the mean distance from i to the other customers in its own cluster and $b(i)$ is the mean distance to the customers of the nearest other cluster. Values close to 1 indicate well-separated clusters.

Profiling and evaluation. For each cluster we compute the average R, F and M, its size and its share of revenue, and give it a descriptive name (for example, loyal high spenders or lapsed customers). Quality is assessed with the silhouette score and by checking that segments are stable across random subsets of the data.

Interactive exploration. Results will be presented in a simple application for exploring customer segments, identifying repeat buyers and visualizing spending patterns. The overall pipeline is shown in Figure 1.



Figure 1: Analysis pipeline for RetailPulse.

3 Dataset Details

We use the *Online Retail II* dataset from the UCI Machine Learning Repository, containing **1,067,371 transactions** from a UK-based online retailer over two years, freely available under the CC BY 4.0 licence. Each row is one line item of an invoice. Table 1 summarizes the dataset. If needed, the analysis can be run on a subset (for example, a single year or a sample of customers) while still giving meaningful results.

Table 1: Summary of the dataset.

Attribute	**Value**
Source	UCI Machine Learning Repository – Online Retail II
Records	1,067,371 transactions
Period covered	Two years
Retailer	UK-based online retailer
Licence	CC BY 4.0
Fields	Invoice, Stock code, Description, Quantity, Invoice date, Unit price, Customer ID, Country